CAMERA MOTION ESTIMATION

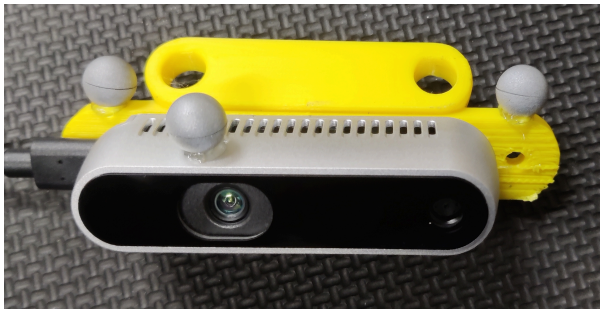
Enhanced MAC RGB-D VIO

Gyro calibration, keyframe tracking and rotation–translation fitting

Method and measured walking results

29 October 2026

Our measurement setup



RealSense camera with a rigid OptiTrack marker cluster.

RGB and depth
30 image pairs per second

Gyroscope
200 angular-rate samples per second

OptiTrack
120 reference poses per second

Depth gives distance in meters. The gyro measures how fast the camera turns.

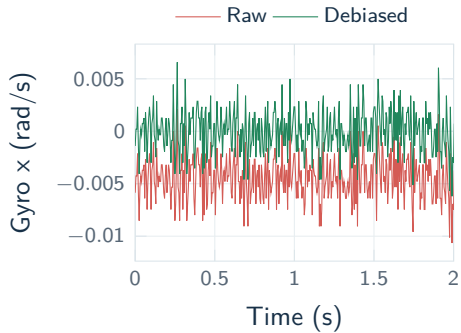
Why errors build up

- A small gyro offset creates tilt over time.
- Short visual steps can be smaller than measurement noise.
- A new rotation needs a new translation fit.

$$\Delta\theta = b_g T$$

b_g: constant angular-rate error.
T: time. $\Delta\theta$: accumulated angle error.

Example: $0.25^\circ/\text{s} \times 135\ \text{s} = 33.75^\circ$.



Actual startup samples from the walking capture.

Camera motion in one equation

$$\boldsymbol{P}_k = \boldsymbol{R}_{kc}\boldsymbol{P}_c + \boldsymbol{t}_{kc}$$

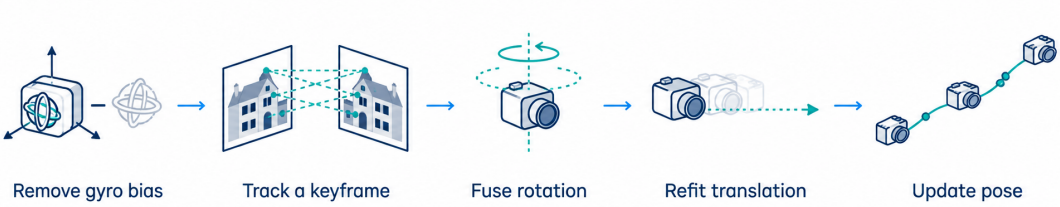
$\boldsymbol{P}_c, \boldsymbol{P}_k$	The same 3D feature in the current and keyframe cameras, in meters.
$\boldsymbol{R}_{kc}$	A 3×3 rotation matrix. It changes the feature's direction.
$\boldsymbol{t}_{kc}$	A 3D translation vector, in meters. It changes the feature's position.

Physical meaning

First rotate a point from the current view into the keyframe view. Then shift it until the two measurements agree.

Camera axes: x right, y down, z forward. The world starts at the first camera pose.

Processing steps



Depth supplies metric translation.

The debiased gyro supports rotation between the same image timestamps.

Concept diagram. All measured results appear later in the presentation.

Gyro bias calibration

$$\boldsymbol{b}_g = \frac{1}{N}\sum_{j=1}^{N} \boldsymbol{\omega}_{\mathrm{raw},j}, \qquad \boldsymbol{\omega}_{\mathrm{corr}} = \boldsymbol{\omega}_{\mathrm{raw}} - \boldsymbol{b}_g$$

N	Number of samples collected while the camera is still.
$\boldsymbol{b}_g$	Mean stationary gyro offset, in rad/s.
$\boldsymbol{\omega}_{\mathrm{raw}}, \boldsymbol{\omega}_{\mathrm{corr}}$	Measured and corrected angular rates, in rad/s.

Physical meaning

Subtract the mean rate measured at rest.

Startup: at least 2 s and 400 samples. Per-axis standard deviation below 0.01 rad/s.

At confirmed stops: $\boldsymbol{b}_g^{+} = \boldsymbol{b}_g + \alpha(\bar{\boldsymbol{\omega}}_{\mathrm{raw}} - \boldsymbol{b}_g)$, with $\alpha = 1 - e^{-\Delta t/(60\,\mathrm{s})}$. The new bias is $\boldsymbol{b}_g^{+}$. The bar means interval average. Δt is elapsed seconds and α is unitless.

Gyro rotation between images

$$\boldsymbol{R}_{\text{imu}} = \prod_{j} \text{Exp}(\bar{\boldsymbol{\omega}}_{\text{corr},j}\, \Delta t_j)$$

- $\bar{\boldsymbol{\omega}}_{\text{corr,j}}$ Mean corrected rate between adjacent IMU samples, in rad/s.
- Δt_j Actual time between those samples, in seconds.
- Exp Converts an axis–angle vector, in radians, into a 3D rotation matrix.
- $\boldsymbol{R}_{\text{imu}}$ Current-to-keyframe rotation from the gyro.

Physical meaning

Multiply the small turns in timestamp order to obtain the total rotation.

Use camera-frame rates and image timestamps. Vision covers missing IMU intervals.

Stationary gating and zero steps

$$d_{\text{flow}} = \text{median}_i \, \|\boldsymbol{u}_{c,i} - \boldsymbol{u}_{p,i}\| < 0.35 \text{ pixels}$$

$$\max_j \|\boldsymbol{\omega}_{\text{corr},j}\| < 0.025 \text{ rad/s}$$

$\boldsymbol{u}_p, \boldsymbol{u}_c$: previous and current feature pixels.
i: valid track. j: gyro sample in this interval.

Confirm with at least 20 tracks over three frames.

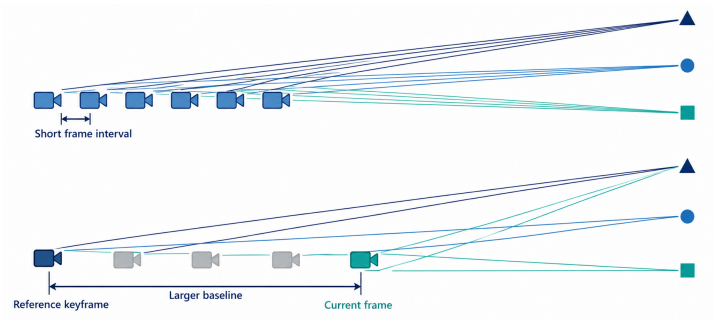
Output: $\Delta \boldsymbol{p}_w = \mathbf{0}$ and $\boldsymbol{v}_w = \mathbf{0}$.

Physical meaning

Small image motion and a small turning rate indicate a stop. Hold the current position.

$\Delta \boldsymbol{p}_w$: world displacement in meters. $\boldsymbol{v}_w$: velocity in m/s. Keep the existing keyframe offset to avoid a backward jump.

Keyframes give a larger visual baseline



New keyframe if any condition holds

Median disparity > 3.5 px

Relative turn $> 2.5°$

Inliers $< 65\%$ of reference points

Physical meaning

A fixed reference lets visual motion grow above the noise.

Disparity is keyframe-to-current pixel motion. Inliers agree with the 3D motion fit.

Depth turns image pixels into 3D points

$$\boldsymbol{P} = d \begin{bmatrix} (u - c_x)/f_x \\ (v - c_y)/f_y \\ 1 \end{bmatrix}$$

- u, v Feature position in the image, in pixels.
- c_x, c_y Camera image center, in pixels.
- f_x, f_y Camera focal lengths, in pixels.
- $d, \boldsymbol{P}$ Depth and the resulting 3D point, in meters.

Physical meaning

The pixel gives direction. Depth gives distance along that direction.

At 2 m depth, one pixel represents about 3.3 mm when $f_x \simeq 607$ px.

Uncertainty sets the weight of each match

$$\Sigma_{3D} = \boldsymbol{J}\Sigma_m \boldsymbol{J}^T$$

$$\Sigma_r = \Sigma_k + \boldsymbol{R}\Sigma_c \boldsymbol{R}^T, \qquad \boldsymbol{\Omega} = \Sigma_r^{-1}$$

- $\Sigma_m, \boldsymbol{J}$ Pixel/depth uncertainty and its conversion to 3D.
- Σ_k, Σ_c Keyframe/current 3D covariances, in m^2.
- $\Sigma_r, \boldsymbol{\Omega}$ Match covariance in m^2 and inverse covariance in m^{-2}.

Physical meaning

Uncertain directions receive less weight. Combine both covariances in the same frame.

$\boldsymbol{J}$ carries the conversion from pixels and depth into 3D position errors.

Stage 1: rotation fusion

$$\phi_f = 0.75\,\mathrm{Log}(\boldsymbol{R}_{\mathrm{vis}}) + 0.25\,\mathrm{Log}(\boldsymbol{R}_{\mathrm{imu}})$$
$$\boldsymbol{R}^* = \mathrm{Exp}(\phi_f)$$

$\boldsymbol{R}_{\mathrm{vis}}$	Rotation fitted from matched 3D points.
$\boldsymbol{R}_{\mathrm{imu}}$	Debiased gyro rotation over the same time interval.
Log, ϕ_f	Matrix-to-axis–angle map and the fused angle vector, in radians.

Physical meaning

Average small rotation vectors, then convert the result to a valid rotation matrix.

The fixed 75/25 split is a design choice. ϕ_f is an angle, not a rate.

Stage 2: translation with rotation fixed

$$\boldsymbol{d}_i = \boldsymbol{P}_{k,i} - \boldsymbol{R}^* \boldsymbol{P}_{c,i}$$

$$\boldsymbol{t}^* = \left(\sum_i w_i \boldsymbol{\Omega}_i \right)^{-1} \sum_i w_i \boldsymbol{\Omega}_i \boldsymbol{d}_i$$

- $\boldsymbol{d}_i$ Shift suggested by match i, in meters, after the fused rotation.
- $w_i, \boldsymbol{\Omega}_i$ Unitless outlier weight and information matrix in m^{-2}.
- $\boldsymbol{t}^*$ Weighted best translation, in meters.

Physical meaning

With the turn fixed, choose the shift that best aligns the points.

Huber weights reduce outlier influence. Repeat the exact weighted solve until stable.

A reference pose anchors the world position

$$\boldsymbol{R}_{wc} = \boldsymbol{R}_{wk}\boldsymbol{R}_{kc}$$

$$\boldsymbol{p}_{wc} = \boldsymbol{p}_{wk} + \boldsymbol{R}_{wk}\boldsymbol{t}_{kc}$$

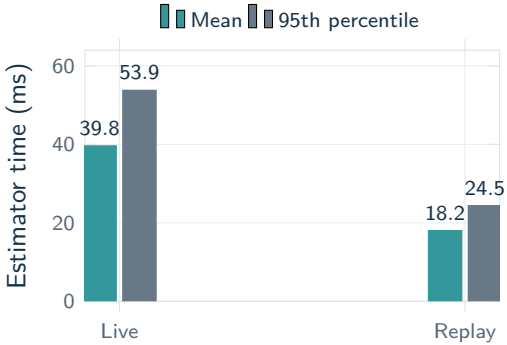
$\boldsymbol{R}_{wk}, \boldsymbol{p}_{wk}$	Keyframe orientation and position in the world.
$\boldsymbol{R}_{kc}, \boldsymbol{t}_{kc}$	Current camera pose relative to that keyframe.
$\boldsymbol{R}_{wc}, \boldsymbol{p}_{wc}$	Current camera orientation and position in the world.

Physical meaning

Rotate the local translation into the world. Then add it to the fixed keyframe position. All position vectors use meters.

Anchor each pose to its keyframe. Never add the full relative offset twice.

CPU work and recording



Less repeated work
Batch RANSAC hypotheses.
Compute uncertainty at feature locations.

Separate file writing
Background workers save raw data.
Explicit counters report lost frames.

Replay timing is offline. It does not establish 30 FPS during live walking. All estimator computation uses the CPU.

Clock synchronization with independent gyro data

$$t_{\mathrm{GT}} = t_{\mathrm{camera}} + \tau$$

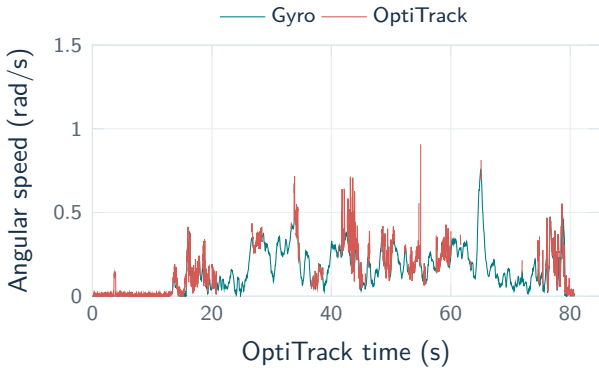
t: time in seconds.

τ: clock offset in seconds.

Compare raw gyro turning with OptiTrack orientation changes.

Correlation: 0.912

40.58 s of reliable support



Use 0.3 s orientation windows to reduce noisy derivatives. No position error is used to choose the time shift.

Accuracy measurement

$$\mathrm{ATE}_{\mathrm{RMSE}} = \sqrt{\frac{1}{M}\sum_{i=1}^{M} \left\| \boldsymbol{R}_a \boldsymbol{p}_i^{\mathrm{est}} + \boldsymbol{t}_a - \boldsymbol{p}_i^{\mathrm{GT}} \right\|^2}$$

M	Number of valid, time-matched position pairs.
$\boldsymbol{R}_a, \boldsymbol{t}_a$	One rigid coordinate alignment. Scale stays equal to one.
$\boldsymbol{p}^{\mathrm{est}}, \boldsymbol{p}^{\mathrm{GT}}$	Estimated and reference positions, in meters.

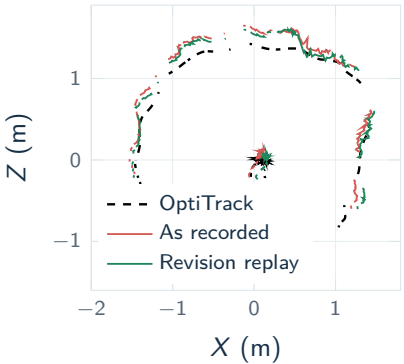
Physical meaning

ATE measures typical position error after rigid alignment, in meters.

902 matched poses cover 63.7% of the recorded trajectory.

Exclude GT gaps above 0.1 s. This take has 105 such gaps, up to 2.84 s long.

Walking trajectory on valid reference intervals



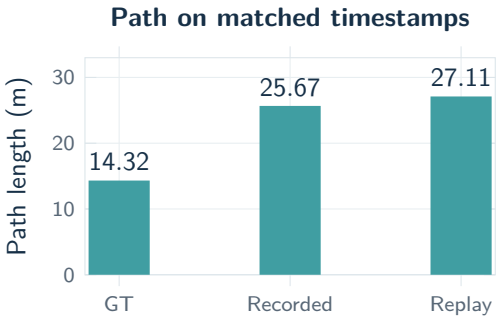
0.420 m
Recorded ATE

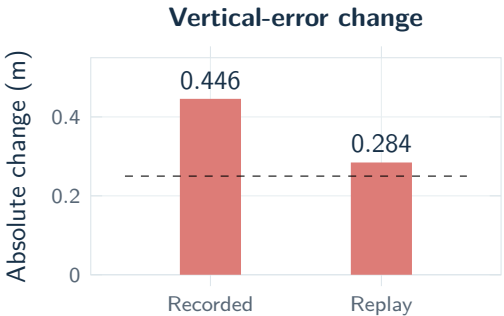
0.308 m
Revised replay ATE

About 27% lower ATE on this valid subset.

The figure breaks lines across missing GT. This take also served to validate the revision, so it is not an independent held-out test.

Path inflation and vertical error remain





Tracking failures: 58 recorded, 66 in the revised replay.

Dashed line: 0.25 m target. Vertical change uses aligned OptiTrack Y between evaluated endpoints. Paths bridge excluded intervals.

What the evidence supports

Established on this capture

The revised replay reduces ATE and estimator CPU time.
Twenty numerical and integration tests pass.

Still unresolved

Path inflation, motion through tracking loss, and vertical error.
Zero-drop 30 FPS during a new live walking trial.

Next measurements

Improve marker visibility and measure the marker-to-camera offset.
Run repeated walking trials with matched OptiTrack recordings.

A short updated stationary live check reached 29.87 FPS with five acquisition drops. It does not establish 30 FPS during walking.